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High-performance one-stage detector for SiC crystal defects based on convolutional neural network

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ABSTRACT

SiC (silicon carbide), as the most important third-generation semiconductor material, has huge market prospects in numerous fields, such as 5G base stations and new energy vehicle charging piles. The identification of SiC crystal defects is essential for improving crystal quality. Currently, this study relies mainly on artificial methods to identify defects, which have significant limitations in terms of accuracy and efficiency. Thus, to quickly detect and classify different SiC crystal defects in complex scenarios, a convolutional neural network-based SiC crystal defect detection (SCDD-Net) model is presented for the first time in this study. SCDD-Net uses an improved online convolutional re-parameterization method that can effectively extract the features of SiC crystal defects and decrease the large training overhead. We devised a new spatial pyramid pooling module that, when combined with the global context block, enables the fast fusion of high-level crystal defects and underlying features. We also designed an anchor-based decoupling detection head network to identify smaller crystal defects. By collecting and processing more than 5300 high-quality microscopic images, we built a fine-grained labeled SiC crystal defect image dataset, SiC-Crystal-5K, for the first time. The experimental results show that the SCDD-Net has excellent detection accuracy compared to other state-of-the-art models. The mean average precision for high-resolution SiC crystal defect identification reached 99.53%, corresponding to a single-image detection speed of 102 fps. In addition to crystal-defect detection, the SCDD-Net model can be used as a general-purpose detector in a wide range of scenarios.

1. Introduction

As a typical representative third-generation semiconductor material, silicon carbide (SiC) has the advantages of a large forbidden bandwidth, high critical breakdown electric field, high thermal conductivity, and high saturation rate. It is widely used to fabricate semiconductor chips and devices [1]. However, due to the different epitaxial growth patterns observed during the growth process of SiC crystals, the generation of crystal defects of different shapes and types is inevitable, which can significantly affect the performance and yield of the final device. This is also the main factor that causes the degradation and failure of electronic devices [2]. Thus, quickly identifying and distinguishing different types of SiC crystal defects is critical for the production of SiC.

Previous crystal defect detection methods were mainly performed in two ways: manual inspection and traditional machine-learning detection. Manual inspection is inefficient, and detection accuracy is not guaranteed. Traditional machine-learning techniques use manual features to classify surface defects [3]. Common manual features include local binary patterns [4] and histograms of oriented gradients [5]. These methods have led to significant improvements in the detection of surface defects. However, manually designed feature windows have several limitations. The window is not targeted in the region selection strategy, and window redundancy is a serious and complicate to implement issue. In addition, it is susceptible to environmental influences such as lighting conditions and background. These algorithms lacked adaptability and robustness. In contrast, deep-learning models can automatically learn features from large amounts of data and perform iterative optimization during the learning process to improve accuracy. Simultaneously, deep-learning models can automatically adapt to different types of perceptual data when dealing with large-scale data and have

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better generalization capabilities, which help them deal with data that have not been seen in training, resulting in better generalization across different scenarios.

Deep-learning-based target detection algorithms are flourishing in the field of computer vision, occupying a critical position in a wide range of applications. These can be divided into two main categories: two-stage and one-stage object detection algorithms. The former involves two processes: localization and extraction of object regions, and classification of images within the proposed regions to obtain the object classes [6]. Common two-stage algorithms include R-CNN [7], Fast R-CNN [8], Faster R-CNN [9], and Mask R-CNN [10]. The latter is a regression task-based method that directly separates specific classes and regresses the boundaries, simplifying the computation process and is faster than two-stage target detection methods. Common one-stage algorithms include YOLO series [11–17], SSD [18], RetinaNet [19], CenterNet [20], and EfficientDet [21]. The use of deep learning to detect high-magnification crystal defect microscopy images has received little attention in previous studies.

Nakazawa [22] proposed a CNN for wafer map pattern classification to classify wafer defect patterns. They used a very small and highly unbalanced dataset that could only train and validate the model with simulated data, resulting in insufficient detection accuracy. Kyeong and Kim [23] proposed a CNN model to classify single- and mixed-defect patterns on the same wafer. However, for defect detection on a large number of wafer maps, this method improves the accuracy while decreasing the detection efficiency, which is ultimately very expensive. Wang et al. [24] used deformable convolutional neural networks to address the problem of classifying wafer defects; however, their results were not satisfactory for the classification of wafers with a mixture of multiple defects. Cheon et al. [25] proposed a CNN model that can extract features from real wafer images and accurately classify input data into different wafer defect classes. After combining the CNN model with the k-nearest neighbor algorithm, the model can classify unknown defect classes. However, its data are extremely small and highly unbalanced, causing the CNN to bias most of the data sample classes.

In addition, for the detection and identification of SiC crystal defects in specialized fields, some applied instrument companies have developed corresponding detection equipment, such as KLA's Candela Series Defect Detection System [26], which can detect and classify a variety of critical defects on compound semiconductor substrates. However, these devices still have shortcomings: First, some of them use manual calibration of features, and then combine it with the algorithm of traditional graphics for detection, which requires professionals with a large amount of a priori knowledge to operate, and the detection efficiency is also low. More importantly, they are extremely expensive.

In this study, a deep-learning-based SiC crystal defect one-stage detection network model (SCDD-Net) is proposed, which can strike a balance between the accuracy and speed required for practical scientific research.

The main contributions of this study are as follows.

(1) A convolutional neural network model (SCDD-Net) based on deep learning was designed by combining deep learning with SiC crystal defect detection. An improved online convolutional reparameterization method was applied to the backbone network, which could effectively extract the SiC crystal defect features and decrease the large training overhead. The SPPFCSPC (a new spatial pyramid pooling module) was designed and combined with a lightweight global context (GCNet) structure to achieve a fast fusion of high-level crystal defect features and underlying features. An anchor-based decoupling detection head network was devised to improve the efficiency of detecting smaller defects.

(2) Because it is difficult to obtain SiC crystal defect image data publicly, an annotation paradigm was developed with experts in related fields to build our own dataset, SiC-Crystal-5K, with nearly 5300 fine-grained annotated high-resolution crystal defect microscopic images covering 50–200 times.

(3) Comparative experiments were conducted using different high-performance one-stage target detection models. The results showed that the SCDD-Net exhibited excellent performance. The model can be widely used for the rapid identification and localization of industrial-grade SiC crystal defects of different sizes. It can be applied to a wide range of scenarios as a general-purpose target detector.

2. Materials

2.1. SiC crystal defect dataset

A SiC crystal is composed of a substrate and an epitaxial layer, as shown in Fig. 1. Dislocation defects in SiC crystals are classified into four categories: thread-screw dislocations (TSDs), thread-edge dislocations (TEDs), basal plane defects (BPDs), and micropipes (MPs) [27]. During the epitaxial growth of SiC crystals, approximately 98% of the TSDs in the substrate extended to the surface, approximately 95% of the BPDs were converted into TEDs, and a small number of BPDs remained, as shown in Fig. 1. Among the four types of defects, both TSDs and TEDs grew from the substrate to the crystal surface, creating small pit-like features. TEDs have densities of approximately 8000–10,000/cm^2, which is almost 10 times that of the TSDs. BPDs rarely appear on the surfaces of SiC crystals and are typically concentrated on the substrate at a density of 1500/cm^2. The diameter of the MPs varies from one micron to several tens of microns, and they exhibit huge pit-like features on the surfaces of SiC crystals with a density of approximately 0.1–1/cm^2. Consequently, the types and quantities of defects in different crystals cannot be controlled, and traditional recognition methods are difficult and inefficient.

Because SiC crystal defect images are unique and there is no publicly available dataset, the experiments in this study used a dedicated dataset established independently. We used an Olympus BX53 (LED) microscope to observe the 4-inch SiC crystals after completing the etching process and acquired images of different types of SiC crystal defects, covering a magnification range of 50×–200×, thereby largely increasing the diversity of the image dataset. Fig. 2 shows the optical micrographs of the substrate dislocation defects in SiC after selective etching. Fig. 3 shows the optical microscopy morphologies of some of the observed defects. In total, nearly 530 raw images of crystal defects were acquired, including 100 images at 50×–100×, 430 images at 100×–200×, with resolutions of 1296 × 1024 and 3072 × 2048, respectively.

2.2. Annotation and enhancement of dataset

The Labelme annotation tool was used to annotate the raw image dataset manually. Although the dataset established above has a certain quantity, to ensure the accuracy and robustness of the model under different situations and to avoid the overfitting problem of training owing to the small amount of data [28], we expanded and enhanced the original image data. Examples of data enhancement are shown in Fig. 4.

The main method is flip enhancement that includes flipping the images horizontally, vertically, and mirroring. Gaussian blur/noise enhancement and salt-and-pepper noise can effectively attenuate the effects of high-frequency feature distortion. The other methods are brightness and sharpness enhancement, and global histogram equalization enhancement.

During data enhancement, annotation information was converted accordingly. After augmentation, invalid enhanced images

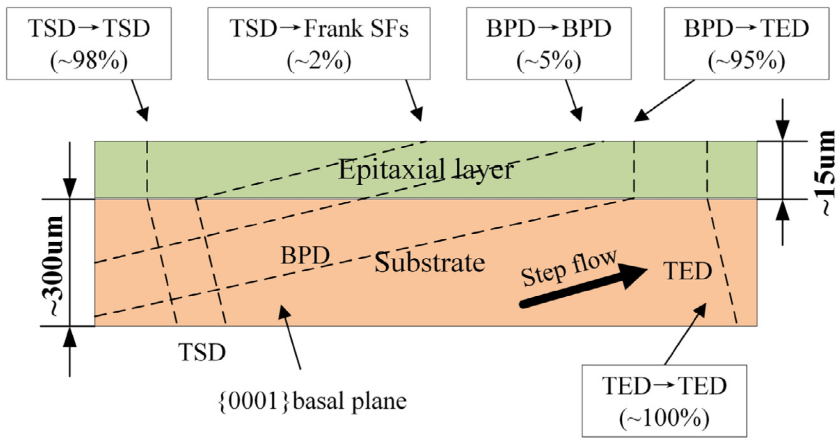

Fig. 1. Structure of SiC crystal and correlation of defects in the growth process.

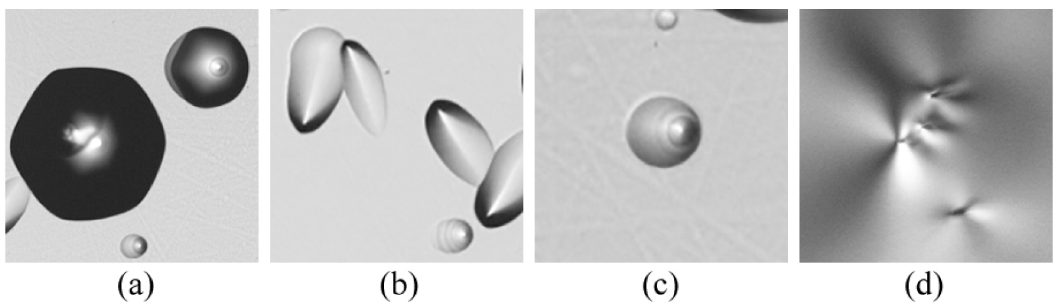

Fig. 2. Four types of SiC crystal substrate defects: (a) TSDs, (b) BPDs, (c) TEDs, (d) MPs.

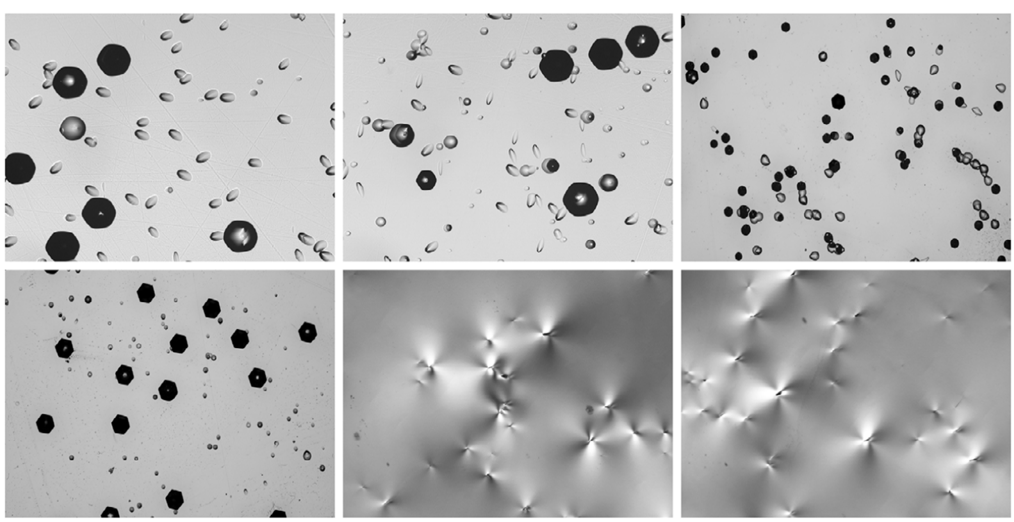

Fig. 3. Some SiC crystal defect images at different magnifications.

were identified and discarded. A dedicated dataset of nearly 5300 images was developed, which was called "SiC-crystal-5k". Information on the original images and the augmented dataset is described in detail in Table 1. In addition, for better training, the dataset was randomly divided into training and test data in a ratio of 8:2.

3. Proposed method

In this study, a SiC crystal defect detection model, SCDD-Net, based on a CNN is proposed. The flowchart of the SCDD-Net algorithm is shown in Fig. 5, and the overall structure of the SCDD-Net is shown in Fig. 6. The backbone network of the proposed SCDD-Net model was improved and designed based on OREPA-VGG, which can efficiently extract SiC crystal defect features and decrease training overhead. In the neck network, the SPPFCSPC module was designed and combined with a lightweight GCNet structure to better integrate the high- and bottom-level features. Finally, an anchor-based decoupling detection head suitable for detecting smaller defects was designed. SCDD-Net exhibits excellent performance and can be widely used for industrial-grade SiC crystal defect detection. This section discusses the detailed design of the network structure.

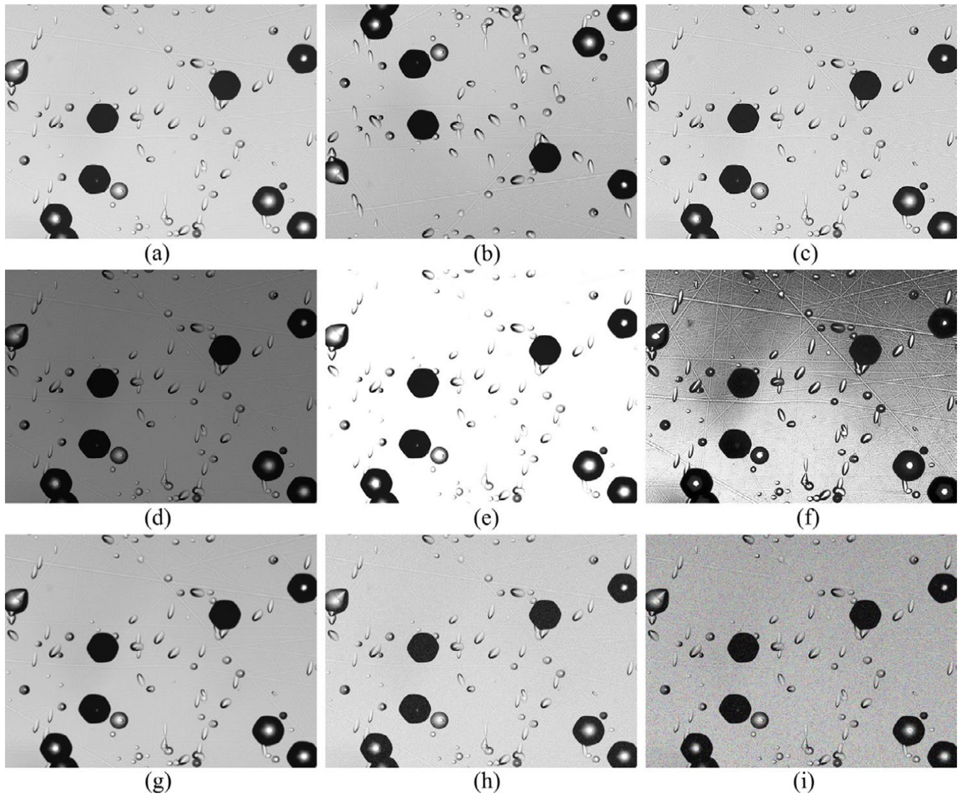

Fig. 4. Examples of data enhancement (a) Original image of SiC, (b) Flip, (c) Sharpness enhancement, (d) Darker, (e) Brighter, (f) Global histogram equalization enhancement, (g) Gaussian blur, (h) Gaussian noise, (i) Salt-and pepper noise.

Table 1
Information of original images and augmented dataset.

Dataset	Original image size	Magnification	Number of images (Original)	Number of images (augmentation)	Actual number of images
SiC-Crystal-5K	3072 × 2048	50×–100×	100	1000	894
		100×–200×	230	2300	2238
	1296 × 1024	100×–200×	200	2000	1993
Total			530	5300	5125

3.1. Design of backbone

Recently, transformers have been used successfully in a range of computer vision tasks, outperforming CNNs in several major domains. However, this comes at the cost of increased computational overhead and parameters. Without transfer learning, a transformer requires extremely large number of data samples for training to obtain better results. Thus, the transformer structure was not used in the backbone design in our study, but in the improved OREPA-VGG [29] structure. OREPA-VGG is a novel structure that combines online convolutional re-parameterization (OREPA) and RepVGG [30]. RepVGG uses only a 3 × 3 convolution and ReLU activation function that is fast, flexible, and memory-economical, without any branching architecture, such as plain or feed-forward.

Structural re-parameterization uses both multibranch model training and single-path model inference, which can equivalently convert the structural parameters during the training time to a different set of parameters during the inference time, thus realizing interconversion between structures. Although the inference process is efficient, it relies heavily on complex blocks of training time to achieve high accuracy, resulting in significant additional training costs. The OREPA structure provides a better solution for these problems. The structural design of OREPA can be seen as an improvement over the diverse branch block (DBB) [31], which is an inception-like unit that complicates the original training model with a heavy parameter trick to ensure that the training model has a better feature representation but transforms the complication into the original model during the inference phase. This allowed the model to achieve excellent feature representation without increasing the inference time. To merge all the components in the block during the training phase, the nonlinear normalization layer must be removed. OREPA removes the batch normalization (BN) layer of the DBB and introduces a linear scaling layer with an effect similar to that of the normalization layer to prevent significant degradation in network performance. The online repair block is illustrated in Fig. 7. Thus, once the linearization phase is completed, the components can be compressed directly into a single 3 × 3 OREPA Conv.

Specifically, a convolutional kernel of size $K_H \times K_W$ is first defined, with the input and output channels denoted as C_{in} and C_{out}, respectively. $X \in \mathbb{R}^{C_{in}\times H\times W}$ and $Y \in \mathbb{R}^{C_{out}\times H'\times W'}$ are the input and output tensors, respectively. We follow the convention of omitting the bias term and denote the convolution as

$$Y = W * X \tag{1}$$

Next, the sequential structure is simplified by considering the stacked representation of the convolutional layers as

$$Y = W_N(W_{N-1} * \cdots (W_2 * (W_1 * X))), \tag{2}$$

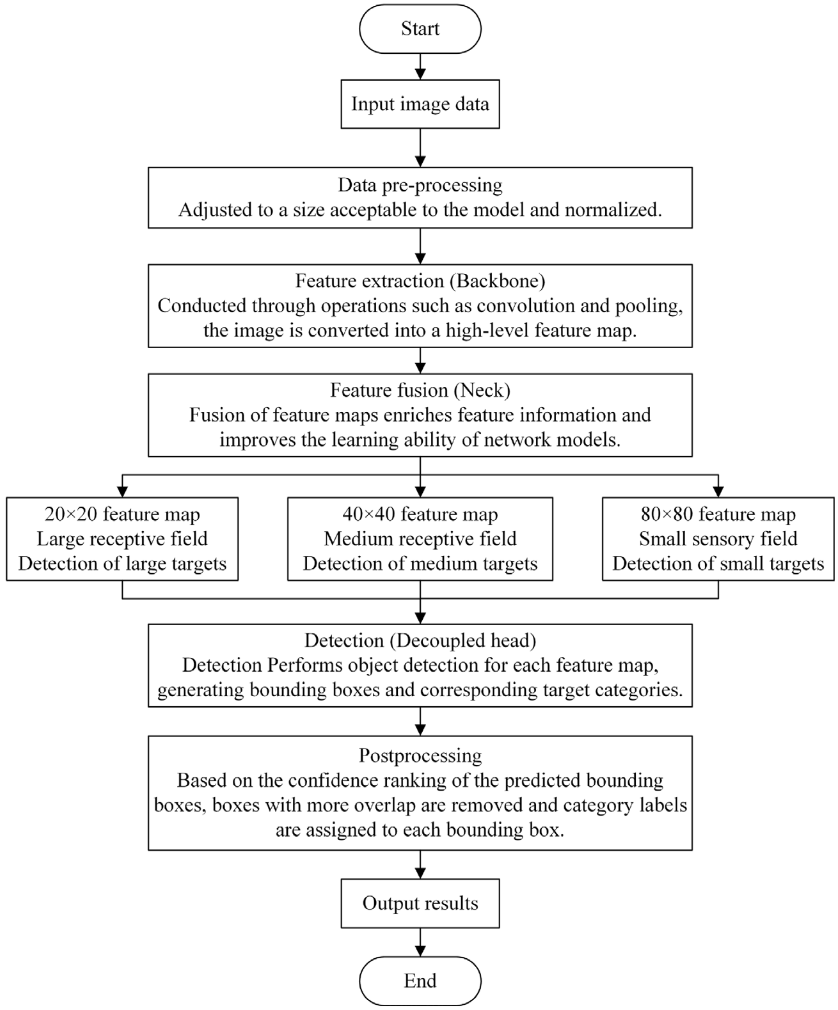

Fig. 5. Flowchart of SCDD-Net algorithm.

where $Wj \in \mathbb{R}^{C_j \times C_{j-1} \times K_{H_j} \times K_{W_j}}$ satisfies $C_0 = C_{in}$, $C_N = C_{out}$. According to the associative law, the kernel is convolved according to Eq. (3). The layers were compressed into a single layer.

$$Y = (W_N\,(W_{N-1} * \cdots W_2 * W_1)) * X = W_e * X, \tag{3}$$

where W_j denotes the weight of the j^{th} layer and W_e denotes the end-to-end mapping matrix.

Finally, the parallel structure is simplified, where multiple branches can be merged into a single branch based on the linear relationship of the convolution, according to Eq. (4). When merging kernels are of different sizes, their spatial centers must be aligned.

$$Y = \sum_{m=0}^{M-1}(W_m * X) = (\sum_{m=0}^{M-1} W_m) * X, \tag{4}$$

where W_m is the weight of the m^{th} branch and $\sum_{m=0}^{M-1} W_m$ is the unified weight.

We have further improved OREPA-VGG in two main ways: first, the frequency prior filter was removed from the original structure, and experiments showed that the improvement in network performance was negligible when using this method on the SiC-Crystal-5k dataset; therefore, only global average pooling (GAP) was used in the network. Next, the depthwise overparameterized convolution (DO-Conv) [32] was adopted to replace the linear depthwise separable convolution (DWConv) [33] with the nonlinear activation layer removed from the original network. The improved structure of OREPA-VGG is shown in Fig. 8.

DO-Conv is very similar to DW SepConv in that it first performs depthwise convolution and outputs an intermediate variable with a number of channels, followed by traditional convolution with a convolution kernel size. However, the difference is that DO-Conv aims to accelerate training with more parameters to improve performance. DO-Conv comprises two linear operations next to each other, and the two operations can be combined into a conventional convolution during inference without any linear processing.

In particular, the DO-Conv operator has two mathematically equivalent expressions, as shown in Eqs. (5)–(7).

$$Z = (D, W) \odot P \tag{5}$$

$$= W * (D \bullet P) \tag{6}$$

$$= (D^T \bullet W) * P, \tag{7}$$

where $\odot$ denotes the output of the DO-Conv operator, D is the depthwise convolution with the trainable kernel $D \in R^{(M \times N) \times D_{mul} \times C_{in}}$, and W is the conventional convolution with the trainable kernel $W \in R^{C_{out} \times D_{mul} \times C_{in}}$, $D_{mul} \geq M \times N$. $P \in R^{(M \times N) \times C_{in}}$ is the input patch, M and N are the spatial dimensions of the patch, and C_{in} is the number of channels in the input feature map. C_{out} is the number of channels in the output feature map.

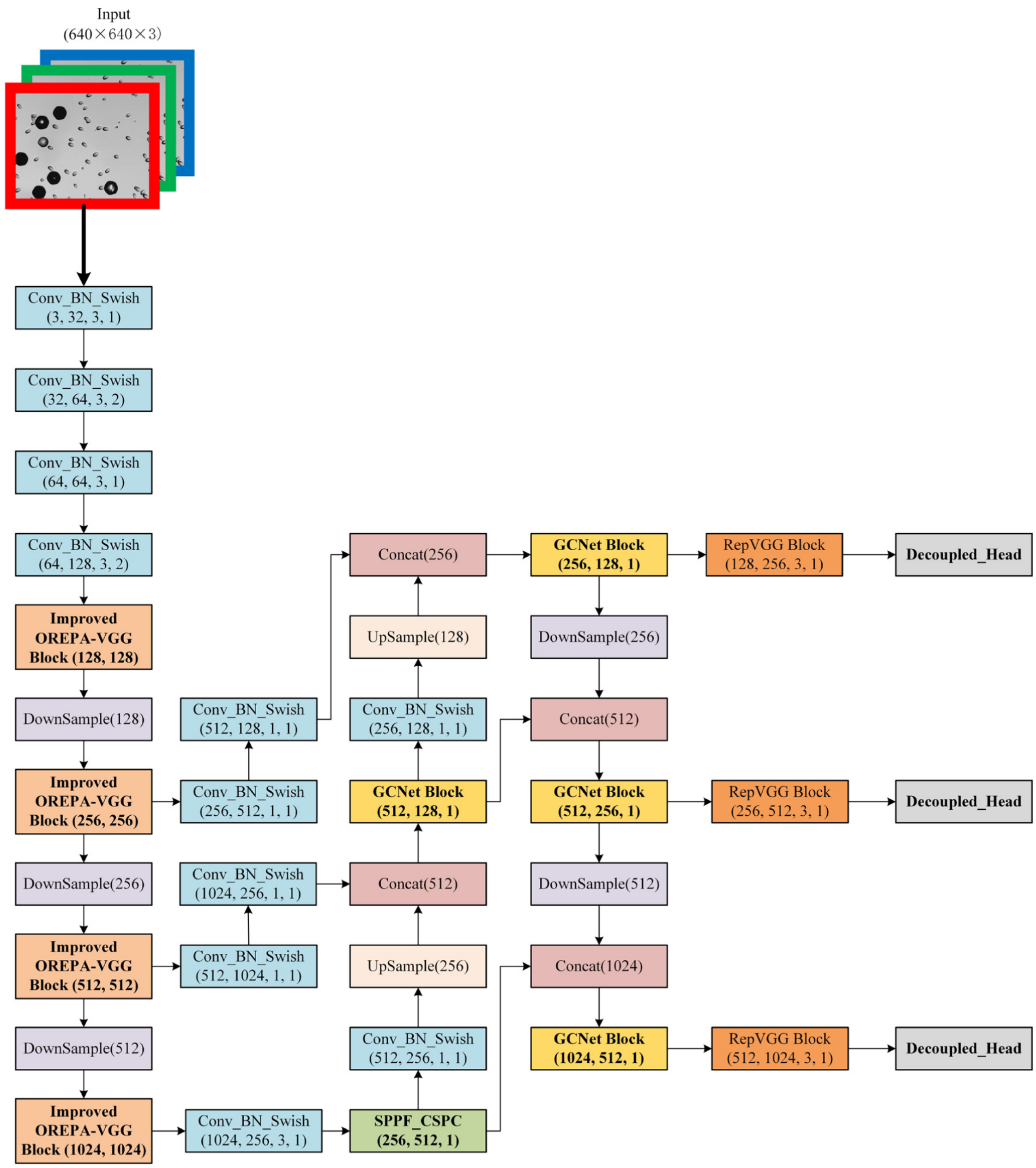

Fig. 6. Architecture of SCDD-Net.

In Eq. (6), $*$ denotes the conventional convolution operator, and $\bullet$ denotes the depthwise convolution operator. The depthwise convolution operator first applies kernel weights D to patch P, obtaining a transformed feature $P' = D \bullet P$, and finally applies kernels W to P' via a conventional convolution operator, yielding $Z = W * P' = W * (D \bullet P)$.

In Eq. (7), $D^T \in R^{D_{mul} \times (M \times N) \times C_{in}}$ is the transpose of $D \in R^{(M \times N) \times D_{mul} \times C_{in}}$ along the first and second axes. The depth-wise convolution operator first uses the trainable kernel D^T to transform W, that is $W' = D^T \bullet W$, and then applies a conventional convolution operator to W' and P, yielding $Z = W' * P = (D^T \bullet W) * P$.

3.2. Design of neck

The most important function of the neck is to perform feature fusion. This idea was first proposed in feature pyramid networks (FPNs) [34], where high-level features are fused with the underlying features, exploiting the high resolution of the underlying features and the rich semantic information of the high-level features to perform independent predictions of multiscale features, which has significantly improved the detection of small objects. The designed neck network contains a spatial pyramid pooling (SPP) [35] and an optimized path aggregation network (PAN) module [36].

First, the SPPFCSPC module was designed for the SPP structure, which is an improvement of the SPPCSPC. The SPPCSPC module was proposed in YOLOv7, which adds Concat operations to the original SPP module to fuse with previous feature maps to obtain richer feature information; however, it also generates a higher number of parameters and requires more computational effort. We were inspired by spatial pyramid pooling-fast (SPPF) [37], which serializes pooling layers with a kernel size of 5. The effect of such an operation is the same as that of the SPP, but with fewer repetitions and a faster operation. Therefore, we optimized

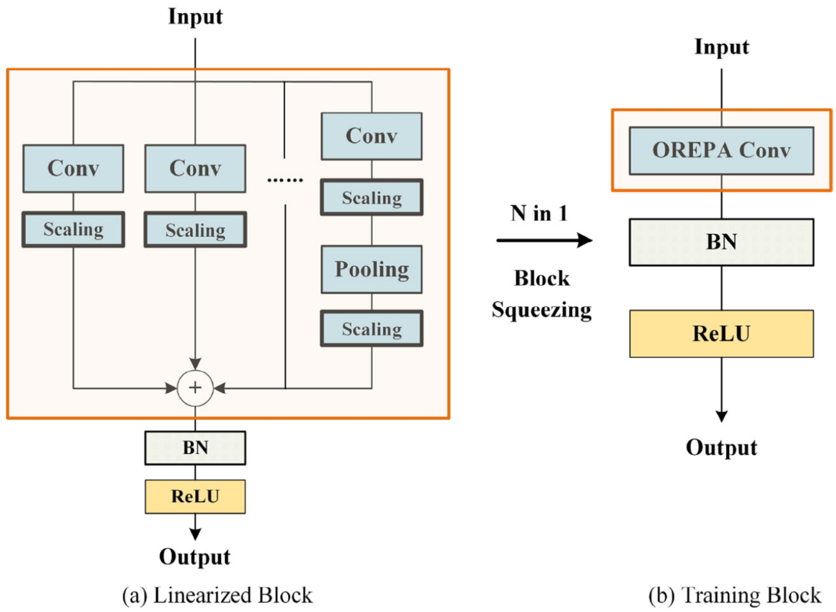

Fig. 7. Architecture of online repair block.

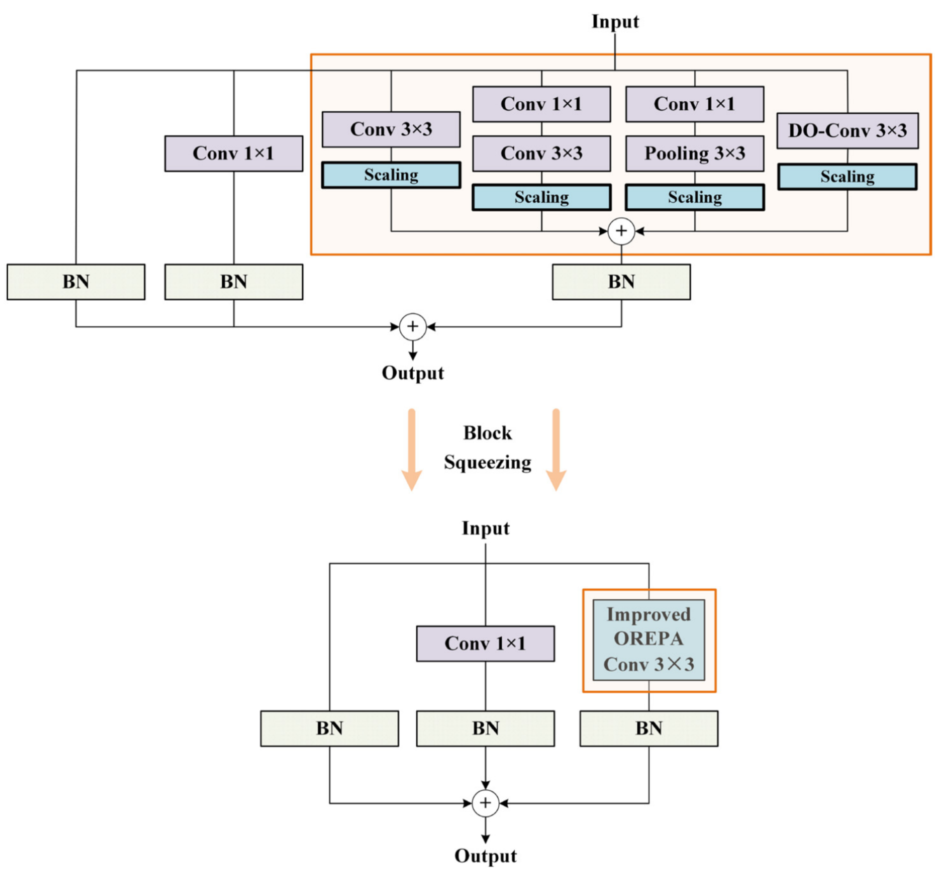

Fig. 8. Architecture of improved OREPA-VGG.

the SPPCSPC based on SPPF to obtain the SPPFCSPC module that can achieve a good speedup while keeping the receptive field unchanged. The SPPFCSPC module is shown in Fig. 9.

Subsequently, for the optimized PAN module, we introduce a global context block (GCNet) [38] for the first time. The structure of the GCNet is shown in Fig. 10. GCNet refers to a nonlocal network (NLNet) [39] and Squeeze-Excitation Network (SENet) [40]. A self-attentive mechanism is used in NLNet to model pixel-pair relationships. However, each location is assigned a location-independent attention map, which results in a large computational load. SENet uses global context to rescale the weights of different channels. However, this approach did not fully exploit the global context. GCNet is a three-step generic framework that unifies NLNet and SENet into global context

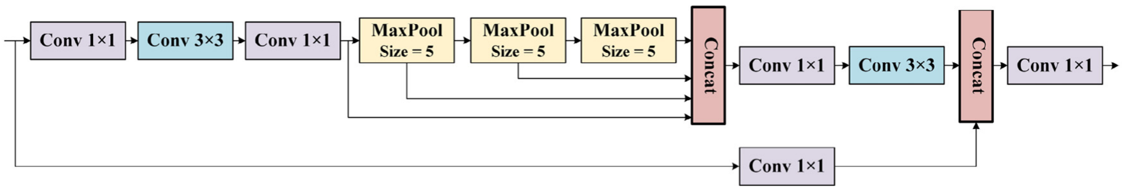

Fig. 9. Architecture of SPPFCSPC module.

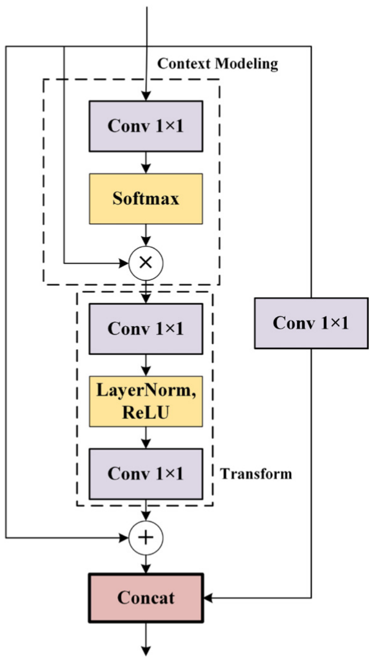

Fig. 10. Architecture of GCNet.

modeling, which is effective in modeling the global context and is sufficiently light.

GC block is represented as:

$$G_i = Z_i + X_{n2}\mathrm{ReLU}\left(LN\left(X_{n1}\sum_{j=1}^{p_F}\frac{e^{X_k \times j}}{\sum_{q=1}^{P_F} e^{X_k \times q}} Z_j\right)\right), \tag{8}$$

where Z_i denotes the feature map input at position i, Z_j denotes the feature map enumerating all possible positions, X_n and X_k denote the linear transformation matrix, P_F denotes the number of positions of the feature map, $\sum_j \alpha_j Z_j$ denotes the context modeling module, $\alpha_j = e^{X_k \times j} / \sum_q e^{X_k \times q}$ is the weight of global attention polling, and $\delta(\bullet) = X_{n2}\mathrm{ReLU}\,(LN\,(X_{n1}\,(\bullet)))$ is the bottleneck transform capturing interchannel dependencies.

The GC block is divided into three steps: (a) global attention pooling, (b) bottleneck transform, and (c) broadcast elementwise addition. It is noteworthy that the 1 × 1 convolution in the transform stage is replaced by a bottleneck transform module that can significantly reduce the number of parameters. Moreover, layer normalization (LN) is performed before ReLU, which reduces the optimization difficulty of the two-layer bottleneck transformation and improves the generalization ability.

3.3. Design of head

After completing the image feature fusion, the neck passes the feature maps at different scales to the head for processing. The head network design was based on the YOLOv7-head structure. After receiving the image information of the three scales from the PAFPN output, the number of channels was adjusted using the RepVGG block and finally predicted using a 1 × 1 convolution. The original head used a coupled head that was implemented by combining classification and regression. However, a conflicting relationship exists between classification and regression in target detection, which can be resolved by decoupling the detection head. Thus, a decoupling method was used in our head network design to improve the 1 × 1 convolution after the RepVGG block. Specifically, the classification and regression were decoupled and divided into two branches, which were first processed using the 1 × 1 convolution for dimensionality reduction, then divided into two 3 × 3 convolution branches for feature transformation, and finally passed through the 1 × 1 convolution layer to output classification and regression. The classification detection head was designed to determine the type of object contained in each feature point. The regression detection head was designed to determine the regression parameters and contained objects for each feature point, and an IoU branch was added to the regression branch. This effectively avoided conflicts between tasks and improved the convergence speed of the training. The structure of the head is shown in Fig. 11. In particular, YOLOX also uses the decoupling detection head; however, its decoupling detection head uses anchor-free, whereas our improved decoupling detection head still uses anchor-based, which has the advantage of being more suitable for the detection of targets with smaller defects.

4. Experiment

4.1. Training environment and details

The following hardware environments were used: Olympus BX53 (LED) microscope, Intel Core i7-12700KF @3.6 GHz CPU, NVIDIA GeForce RTX 3080Ti GPU, and 32 GB DDR4 RAM.

The following software environments were used: Windows 11 64-bit operating system, PyTorch 1.11, Python 3.9, PyCharm IDE, OpenCV 4.5, CUDA 11.5, and CuDNN 8.3.2.

The initialization parameters of SCDD-Net are listed in Table 2.

A stochastic gradient descent method with momentum is applied to modify these parameters. The momentum used to accelerate model convergence was set to 0.937 in the optimization function. After several experiments with different learning rates, the initial learning rate was set to 1e−2 and the weight decay of the optimizer was set to 0.0005. Considering the memory limitations and the impact of the batch on the detection performance, the batch size was set to 16. During the training process, data enhancement techniques were used, such as mosaics, mixup, and random HSV enhancement. For the loss function, the target confidence and classification loss were calculated using binary cross-entropy (BCE) and sigmoid, whereas the coordinate loss was calculated using complete-IoU (CIoU). For label assignment, the simOTA label-matching strategy was used, which effectively decreased the training time required for optimal transport assignment (OTA) while maintaining performance.

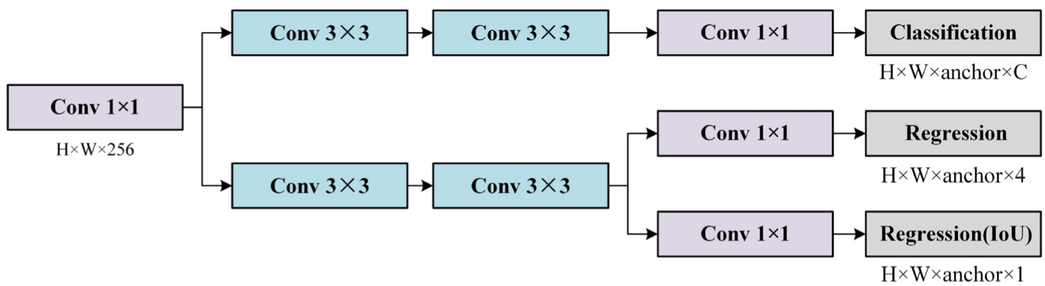

Fig. 11. Architecture of the decoupled head.

Table 2
Initialization parameters of SCDD-Net.

Input size	Gradient descent	Momentum	Learning rate	Weight decay	Batch size	Epoch
640 × 640	SGD	0.937	1e−2	0.0005	16	300

4.2. Performance evaluation

In industrial scenarios, it is particularly important to determine the accuracy of crystal defect detection. Inaccurate types or locations of detected objects can lead to incorrect judgments and prevent accurate improvements in manufacturing processes. To overcome the highlighted problems, precision, recall, F1-score, average precision (AP), and mean average precision (mAP) were used as evaluation metrics for model performance, while also considering the parameters and size of the evaluated model. The formula is as follows:

$$Precision = True_{Positive}/(True_{Positive} + False_{Positive}) \tag{9}$$

$$Recall = True_{Positive}/(True_{Positive} + False_{Negative}) \tag{10}$$

$$AP = \int_0^1 Precision(Recall)d(Recall) \tag{11}$$

$$mAP = \sum_{i=1}^{K} AP_i \quad / K \tag{12}$$

$$F1 - score = 2 \times (Precision \times Recall)/(Precision + Recall), \tag{13}$$

where AP denotes the average precision for a single category, mAP denotes the average precision for each category, F1-score denotes the harmonic mean of precision and recall, K is the number of detected defect categories, TP denotes the correctly predicted samples, that is, the detected boxes with an intersection-to-merge ratio (IoU) $>$ 0.5, FP denotes the incorrect samples predicted as correct samples, i.e., the detected boxes with IoU $\leq$0.5, FN denotes the correct number of samples predicted as incorrect, that is, ground truths that were not detected, and IoU = 0.

The overall schematic of the SiC crystal defect detection device and the actual application software supplement are shown in Fig. 12.

5. Results and discusses

5.1. Performance comparison of different models

The proposed SCDD-Net was trained and validated. The loss, mAP, precision, recall, P-R, and F1 curves during the training process are shown in Fig. 13, and the detection results are shown in Figs. 14–19.

Table 3
Comparison of the results of SCDD-Net and other one-stage network models.

Model	AP_{TSD}	AP_{BPD}	AP_{TED}	AP_{MP}	mAP@0.5	mAP@0.5:0.95
SSD	0.9803	0.9408	0.8859	0.9807	0.9469	0.6050
RetinaNet	0.9361	0.7897	0.4401	0.9988	0.9361	0.5773
CenterNet	0.9900	0.9754	0.9743	0.9995	0.9848	0.7190
EfficientDet-D1	0.9892	0.8837	0.7466	0.9980	0.9044	0.6560
YOLOv3	0.9838	0.9813	0.9805	0.9189	0.9661	0.6180
YOLOv4	0.9878	0.9152	0.9779	0.9504	0.9578	0.6243
YOLOv5-M(r6.0)	0.9852	0.9830	0.9679	0.9784	0.9786	0.7087
YOLOv5-L(r6.0)	0.9861	0.9829	0.9779	0.9873	0.9841	0.7310
YOLOX-M	0.8981	0.9033	0.8995	0.8819	0.8957	0.6630
YOLOX-L	0.9012	0.9066	0.9053	0.8936	0.9017	0.6879
YOLOv7	0.9892	0.9908	0.9891	0.9893	0.9896	0.7253
SCDD-Net(ours)	**0.9959**	**0.9963**	**0.9957**	**0.9931**	**0.9953**	**0.7320**

As shown in Fig. 13, each evaluation metric of SCDD-Net performed well after 300 epochs of training under the set parameter learning strategy. The loss, mAP, and F1-score curves converged rapidly within the first 20 epochs, reflecting the superiority of SCDD-Net.

The red, pink, orange, and yellow boxes represent the TSD, BPD, TED, and MP labels, respectively, as shown in Figs. 14–19. The number above each label represents the confidence level for predicting the defects in the SiC crystals. The model could accurately identify defects without missing them. Thus, the model has a relatively reliable performance.

To further validate the accuracy and effectiveness of the models, a series of comparative experiments were conducted on other one-stage target detection models for a comprehensive comparison. We selected the SSD, RetinaNet, CenterNet, EfficientDet, and YOLO series of algorithms. The parameter settings of each algorithm were maintained at their default values to achieve optimal performance. The comparison of the results for the SiC-Crystal-5k dataset is presented in Table 3.

The results show that the proposed SCDD-Net exhibits excellent performance in terms of detection accuracy, and its comprehensive detection effect is better than that of the existing state-of-the-art detection models. mAP@0.5 and mAP@0.5:0.95 of SCDD-Net can reach 99.53% and 73.20%, respectively; the mAP@0.5 is 4.84%, 5.92%, 1.05%, 9.09%, 2.92%, 3.75%, 1.76%, 1.12%, 9.96%, 9.63%, and 0.57% better than those of SSD, RetinaNet, CenterNet, EfficientDet, YOLOv3, YOLOv4, YOLOv5-M, YOLOv5-L, YOLOX-M, YOLOX-L, and YOLOv7, respectively. The mAP@0.5:0.95 is 12.70%, 15.47%, 1.30%, 7.6%, 11.40%, 10.77%, 2.33%, 0.1%, 6.9%,

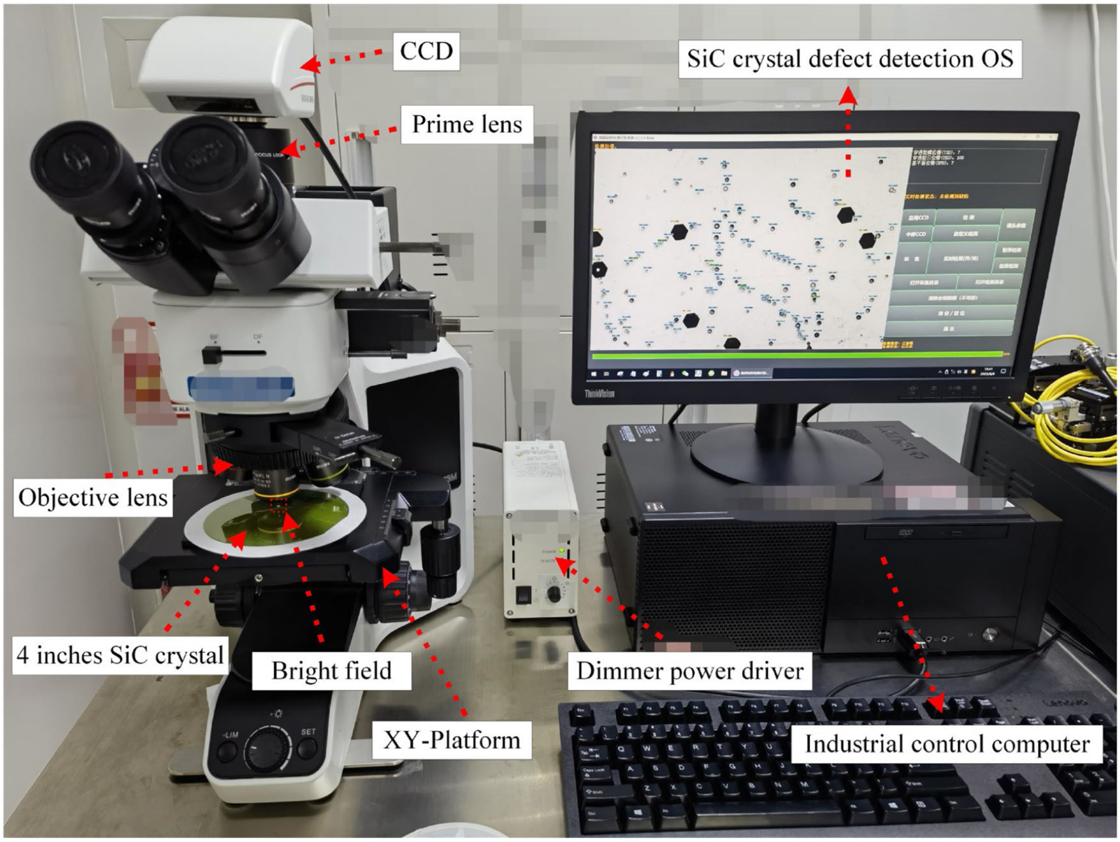

Fig. 12. General schematic of the defect detection device and the actual application software.

4.41%, and 0.67% better than the same algorithms above, respectively.

5.2. Ablation study

To better demonstrate the improvement in the model performance by different contributions, based on the previous section, we conducted ablation experiments to verify the impact. As listed in Table 4, our experiments considered the performance impact of each component in different combinations.

First, we analyzed the effectiveness of the improvement components in the backbone network. We found through experiments that the frequency prior filter (Frep component), which can be understood as a special expression of the pooling layer, has minimal performance improvement. Therefore, we eliminated this component from the original network, which decreased the complexity to some extent.

Second, we found that after using the DO-Conv component, the performance of the model improved by 0.16%, and the FLOPs were decreased from 116.7 G to 107.3 G. This demonstrates the effectiveness of adding depth-separable convolutional operations to conventional convolutional operations, which decreases the FLOPs of the model while improving its performance.

Next, for the neck network, we found that after using our designed SPPFCSPC module instead of SPPCSPC, the model performance improved by up to 0.4% compared to the baseline, whereas there was no additional increase in FLOPs because the SPPFCSPC module accelerated the operation and improved the efficiency of the model while keeping the sensory field unchanged.

After adding the lightweight GCNet structure, we found that the model performance was substantially improved by up to 0.78% compared to the baseline, whereas the FLOPs were directly decreased from 107.3G to 27.4G. We investigated why the improved component was so effective and found that it was due to the fact that GCNet employed bottleneck transformation to decrease redundancy in the global context features, which further decreased the number of parameters and FLOPs. Meanwhile, GCNet utilizes addition to aggregate the global context to all locations, which can better capture remote dependencies and help in network training, thereby substantially improving the network performance.

Finally, for the detection head, although the model FLOPs increased after we adopted the decoupling approach, the performance improvement was up to 1.29% compared to the baseline model and up to 0.51% compared to the traditional coupling head scheme. This is attributed to the fact that the decoupled head extracts the target location and category information separately, learns them separately through different network branches, and finally fuses them. Although the model size increases slightly, it can effectively decrease the computational complexity and enhance the generalization ability and robustness of the model, thus substantially improving the model performance.

5.3. More comparison of different models

Furthermore, based on the previous section, more details of the model are compared, including the number of parameters, FLOPs, and FPS, to better illustrate the performance of the model. The comparison of the details is presented in Table 5.

SCDD-Net uses online structural re-parameterization with different numbers of parameters in training and inference, with a total number of 73.4M parameters. The entire model has a total of 37.7G FLOPs, and a single image input at 640 × 640 image resolution achieves an FPS of 102 on the experimental platform described herein.

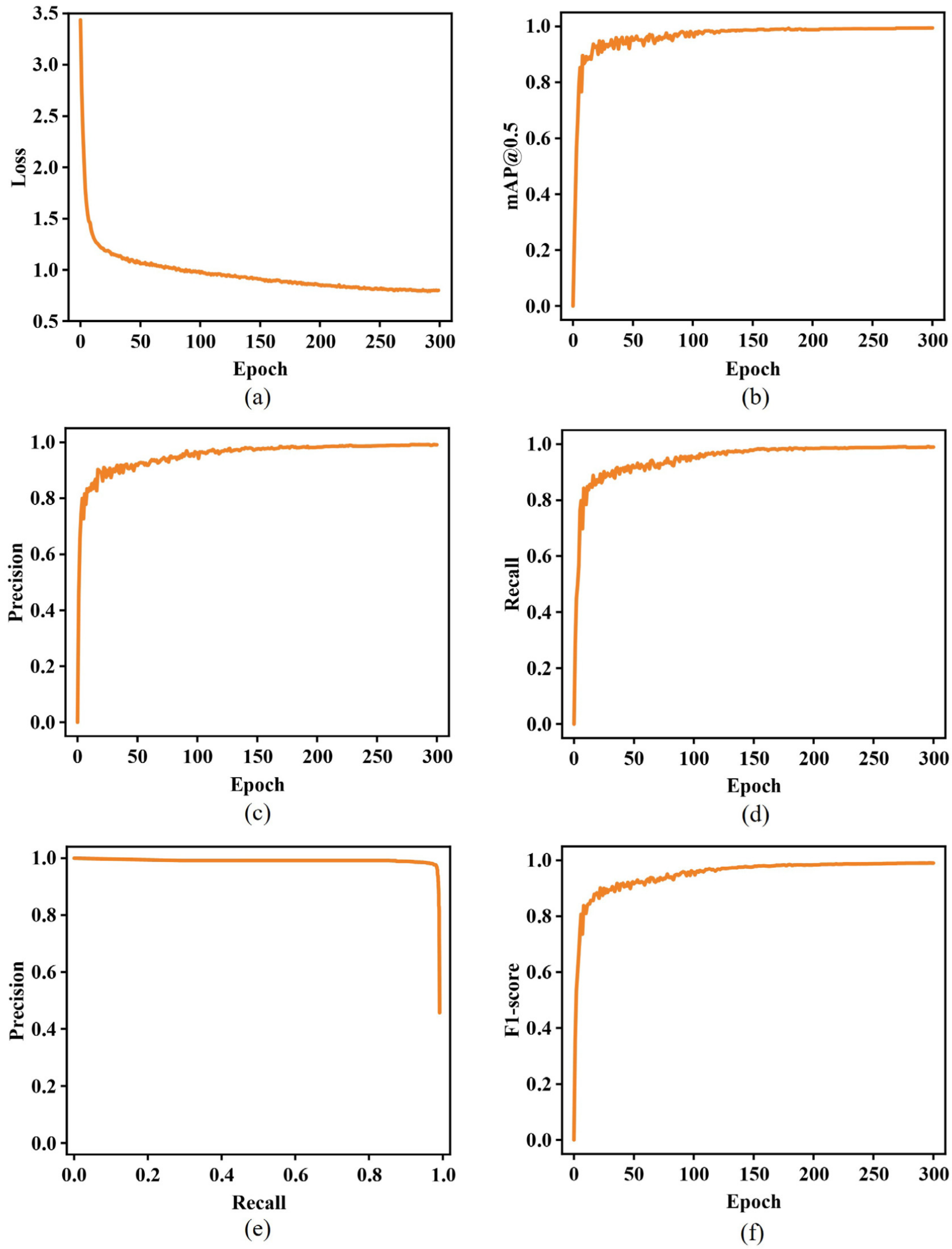

Fig. 13. Curves of (a) loss, (b) mAP, (c) precision, (d) recall, (e) P-R and (f) F1-score during the training process.

Table 4
Results of the ablation study.

Model					mAP@0.5	FLOPs
Baseline (OREPA-VGG + SPPCSPC + ELAN + Coupled head)					0.9824	116.7G
Ablation study						
Backbone		Neck		Head	–	–
Freq	DO-Conv	SPPFCSPC	GCNet	Decoupled		
✗	✗	✗	✗	✗	0.9822	116.7G
✗	✓	✗	✗	✗	0.9840	107.3G
✗	✓	✓	✗	✗	0.9864	107.3G
✗	✓	✓	✓	✗	0.9902	27.4G
✗	✓	✓	✓	✓	0.9953	37.7G

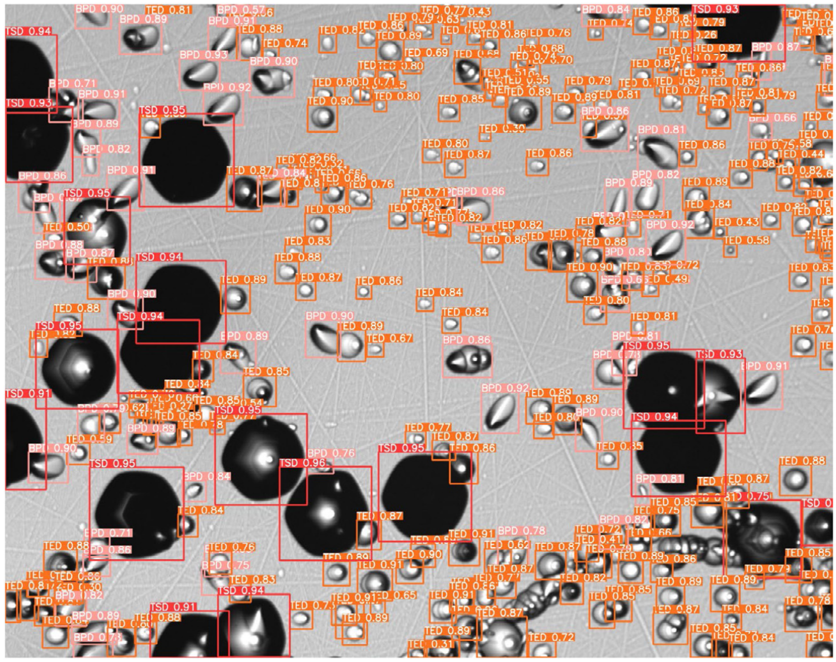

Fig. 14. 200× SiC crystal defect detection result 1.

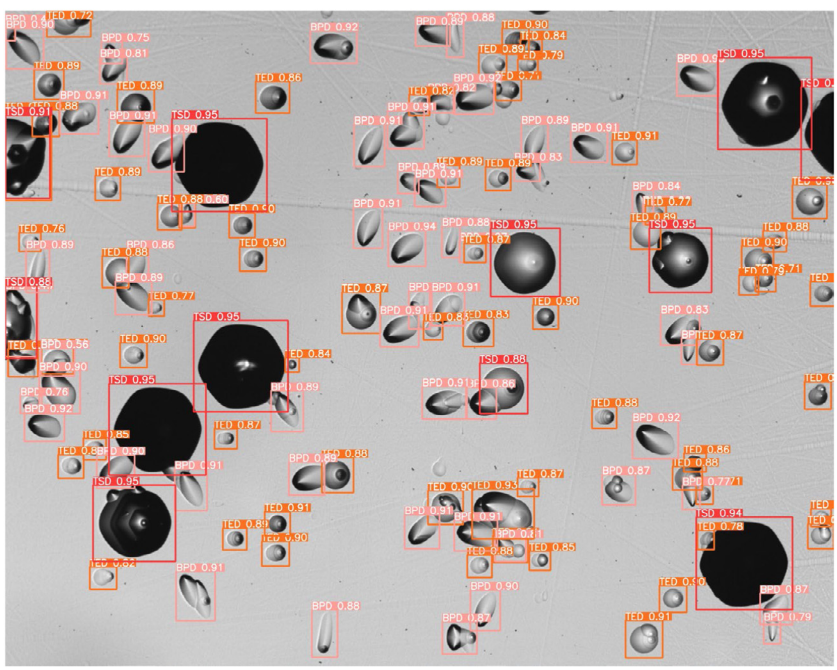

Fig. 15. 200× SiC crystal defect detection result 2.

The results show that compared to other models, the proposed SCDD-Net model achieves higher crystal defect recognition accuracy with a higher FPS and a smaller model size. Although it may not be always the fastest, it has a higher accuracy compared to models with similar FLOPs. It has smaller FLOPs than the models with similar speeds. Thus, the SCDD-Net better balances recognition accuracy with recognition speed.

6. Conclusion

In this study, a CNN-based SiC crystal defect detection model, SCDD-Net, is presented for the first time. In the construction of the network, an improved OREPA-VGG-based backbone network was designed to efficiently extract the features of SiC crystal defects. The SPPFCSPC module was designed and combined with the lightweight GCNet structure, which better integrated high-level features with the underlying features. An anchor-based decoupling detection head can be applied to smaller crystal defects while achieving better performance. Simultaneously, a fine-grained labeled SiC crystal defect image dataset, SiC-Crystal-5K, was established for the first time. The experimental results showed that the SCDD-Net exhibited excellent performance and speed. In the application scenario of SiC defect identification, the average detection accuracy of the method was 99.53%, and

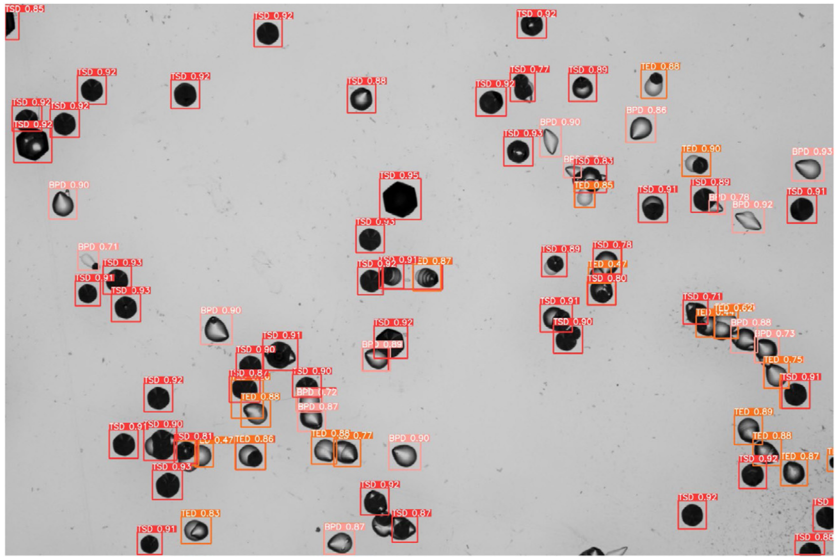

Fig. 16. 100× SiC crystal defect detection result 1.

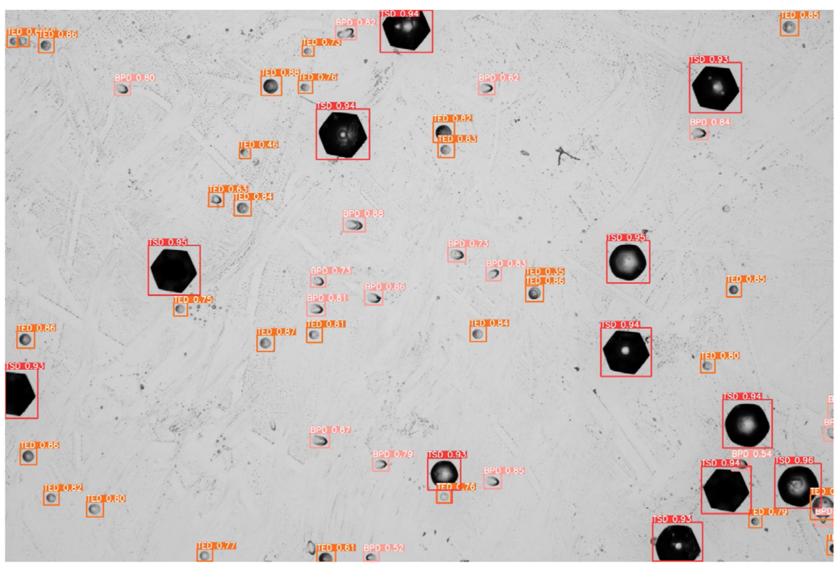

Fig. 17. 100× SiC crystal defect detection result 2.

Table 5
Comparison of SCDD-Net and other network models in more detail.

Model	FLOPs	Parameters	FPS^{3080Ti}
SSD	175.7G	24.0M	112
RetinaNet	146.3G	36.4M	47
CenterNet	70.2G	32.7M	114
EfficientDet-D1	11.5G	6.6M	29
YOLOv3	154.6G	63.0M	66
YOLOv4	141.0G	52.5M	63
YOLOv5-M(r6.0)	47.9G	20.9M	120
YOLOv5-L(r6.0)	107.7G	46.1M	80
YOLOX-M	73.8G	25.3M	91
YOLOX-L	155.6G	54.2M	69
YOLOv7	104.7G	36.5M	108
SCDD-Net(ours)	37.7G	73.4M	102

the real-time detection speed was as high as 102 fps on high-resolution defect images, which was better than those of other state-of-the-art target detection methods. It can be generalized across multiple defect identification scenarios and maintains the stability of the identification accuracy, effectively providing a reference for SiC crystal defect detection techniques.

However, the proposed detection network model still has room for improvement. Usually, structural re-parameterization methods are designed to run on GPUs or dedicated hardware, which is not suitable for low-power devices. Decreasing the overall number of model parameters while maintaining accuracy so that the model can be better applied to low-power devices is a future research direction. We believe that we can solve the problems of parametric quantity and training consumption structurally using convolution with learnable scaling parameters to fuse the branches of the OREPA-VGG training phase. In addition, the dataset will be improved and expanded to include more types of SiC crystal epitaxial layers and substrate defects to better meet actual industrial needs in rich scenarios.

CRediT authorship contribution statement

Haochen Shi: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization.

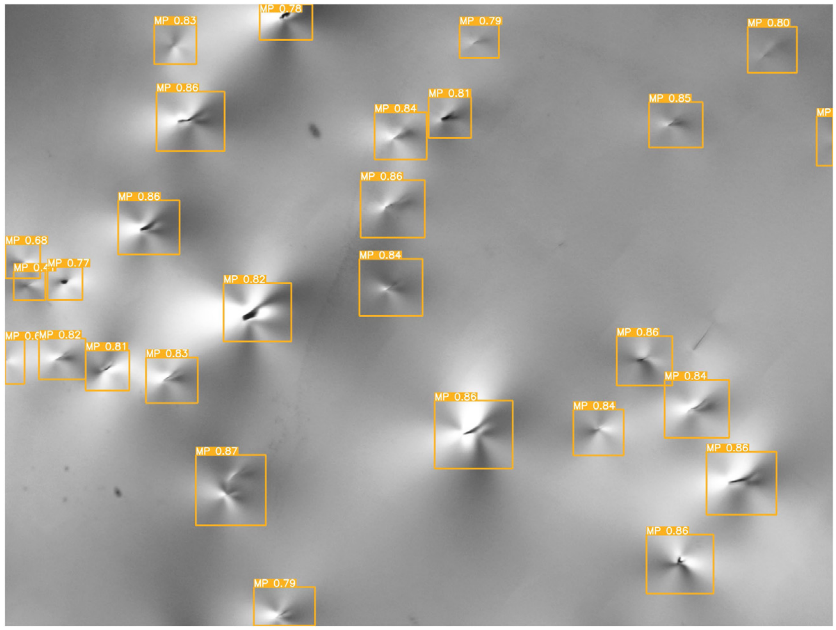

Fig. 18. 50× SiC crystal defect detection result 1.

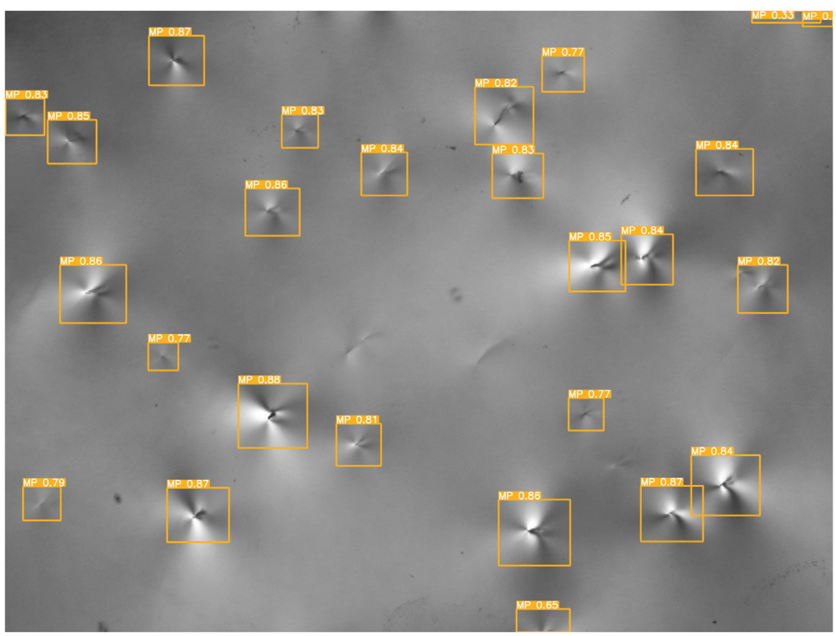

Fig. 19. 50× SiC crystal defect detection result 2.

Zhiyuan Jin: Software, Investigation, Data curation. **Wenjing Tang:** Writing – review & editing, Visualization. **Jing Wang:** Formal analysis. **Kai Jiang:** Investigation. **Mingsheng Xu:** Resources. **Wei Xia:** Writing – review & editing, Project administration. **Xiangang Xu:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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